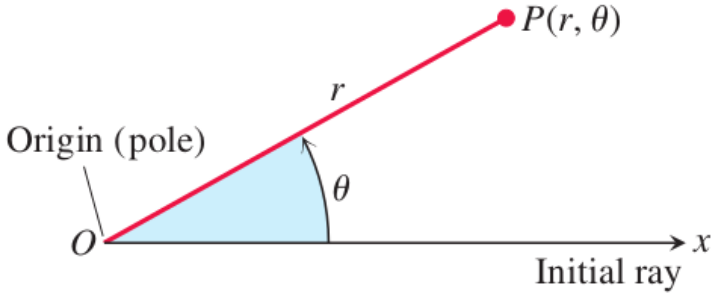
10.3 Polar Coordinates

Definition of Polar Coordinates



Fix an **origin** O **(pole)** and an **initial ray** from O. Then every point P on the plane can be located by assigning to it a **polar coordinate** (r, θ) with

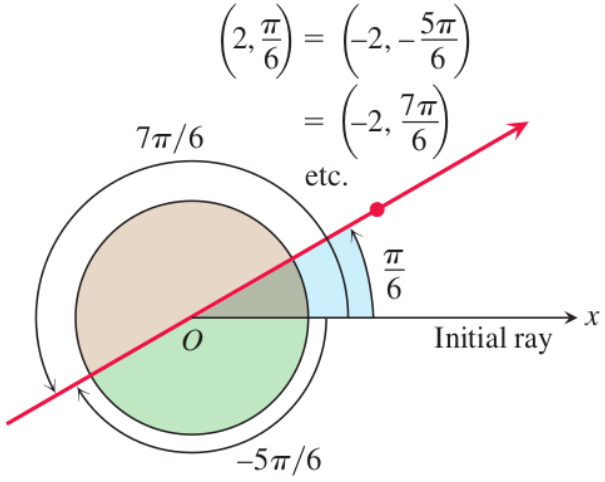
r = the directed distance from O to P,
θ = the angle from initial ray to ray OP.

Polar Coordinates $P = P(r, \theta)$

Here, we note that θ is positive when measured counterclockwise, and negative when measured clockwise.

Sometimes, we also allow r to be negative as well - when we want to indicate that the ray is *going backward*.

Example: Polar coordinates of the point $P(2, \pi/6)$.



Polar Equations and Graphs

A graph can be specified by a polar equation

Example: Recall that in the Cartesian coordinates system, the equation $x = c$ describes a vertical line and the equation $y = d$ describes a horizontal line. In the polar coordinate system, the equations

1. $r = a$ is the circle of radius $|a|$ centred at O
2. $\theta = \theta_0$ is the line through O making an angle θ_0 with the initial ray.

Examples:

1. $r = 1$ and $r = -1$ are the equations representing the same graph, that is, the circle of radius 1 centred at O.
2. $\theta = \pi/6$, $\theta = 7\pi/6$, and $\theta = -5\pi/6$ all represent the same line, the line in the previous example.

We can also define regions using polar equations.

Examples:

1. $1 \leq r \leq 2$ and $0 \leq \theta \leq \frac{\pi}{2}$.
2. $-3 \leq r \leq 2$ and $\theta = \frac{\pi}{4}$.
3. $\frac{2\pi}{3} \leq \theta \leq \frac{5\pi}{6}$ (no restriction on r).

Relating Polar and Cartesian Coordinates

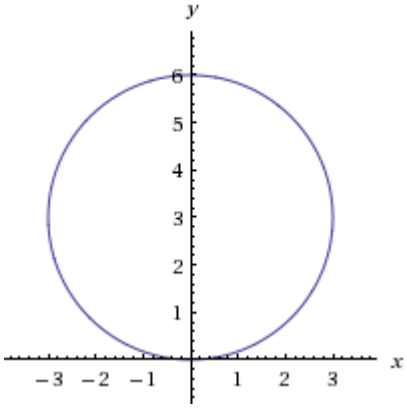
In both systems, we place both the origins together and take the initial polar ray as the positive x-axis. The y-axis is then the ray $\theta = \pi/2,\ r > 0$. Then we will have

$$x = r\cos\theta \qquad y = r\sin\theta \qquad r^2 = x^2 + y^2 \qquad \tan\theta = \frac{y}{x}$$

Example:

Polar equation	Cartesian equivalent
$r\cos\theta = 2$	$x = 2$
$r^2\cos\theta\sin\theta = 4$	$xy = 4$
$r^2\cos^2\theta - r^2\sin^2\theta = 1$	$x^2 - y^2 = 1$
$r = 1 + 2r\cos\theta$	$y^2 - 3x^2 - 4x - 1 = 0$
$r = 1 - \cos\theta$	$x^4 + y^4 + 2x^2y^2 + 2x^3 + 2xy^2 - y^2 = 0$

Example: Find a polar equation for the circle $x^2 + (y - 3)^2 = 9$.



Examples: Find the cartesian equivalent of the following polar equations.

(a). $r\cos\theta = -4$

(b). $r^2 = 4r\cos\theta$

(c). $r = \frac{4}{2\cos\theta - \sin\theta}$